

Short Communication

Papillitis associated with IgLON5 autoimmunity: A novel clinical phenotype

Xiaoyang Li^a, John J. Chen^{a,b,c}, Minjun Hur^{a,b}, Gillian R. Paton^d, Andrew McKeon^{a,c,e}, Anastasia Zekeridou^{a,c,e,*}

^a Department of Neurology, Mayo Clinic, Rochester, USA

^b Department of Ophthalmology, Mayo Clinic, Rochester, USA

^c Center of MS and Autoimmune Neurology, Mayo Clinic, Rochester, USA

^d Casey Eye Institute Division of Neuro-ophthalmology, Oregon Health & Science University, USA

^e Department of Laboratory Medicine and Pathology, Mayo Clinic, Rochester, USA

ARTICLE INFO

Keywords:

IgLON5

Papilledema

Optic nerve

Optic disc edema

Encephalitis

ABSTRACT

Objectives: To describe papillitis as a clinical phenotype of IgLON5 autoimmunity.

Methods: We retrospectively reviewed patients with IgLON5 autoimmunity who had optic neuropathy, optic neuritis, or optic disc edema. Sera from patients with recurrent papillitis were tested for IgLON5 antibodies.

Results: We found two elderly males presenting with papillitis in the presence of IgLON5 antibodies. CSF pleocytosis was present and partial vision improvement occurred in one patient despite immunotherapy. Sera from 18 patients with recurrent papillitis were negative for IgLON5 antibodies.

Conclusion: Papillitis could be a manifestation of IgLON5 disease, with or without accompanying cognitive, sleep, and movement disorders.

1. Introduction

IgLON5 autoimmunity commonly affects the elderly population and may present insidiously posing diagnostic challenges (Gaig et al., 2017; Gelpi et al., 2016; Honorat et al., 2017; Sabater et al., 2014). The clinical presentation is heterogeneous with phenotypes including cognitive impairment, sleep disturbance, gait/movement disorders, bulbar dysfunction, and motor neuron disease-like syndrome (Gaig et al., 2017; Grüter et al., 2023; Honorat et al., 2017; Sista et al., 2022). Early recognition and immunotherapy within the first year of disease presentation are associated with better outcomes (Grüter et al., 2023). Oculomotor involvement such as supranuclear gaze palsy has been reported, however, optic nerve involvement is not an established clinical presentation in this disease. We report two cases of papillitis associated with IgLON5 autoimmunity, suggesting a possible new clinical phenotype in this rare disorder.

2. Methods

This study was approved by the Mayo Clinic IRB; patients gave

consent for the use of their medical records for research. We retrospectively identified patients with IgLON5 autoimmunity (defined as the presence of IgLON5-IgG in serum and/or CSF, tested with both a tissue-based and cell-based indirect immunofluorescence assay, as previously described) (Honorat et al., 2017) and optic neuropathy. We also tested 18 patients' sera with recurrent papillitis for IgLON5-IgG.

3. Results

Two patients were identified meeting the inclusion criteria of IgLON5-IgG positivity and optic neuropathy. All sera tested from the 18 patients with recurrent papillitis were negative for IgLON5-IgG.

3.1. Patient 1

A man in his late 60s, with a prior history of pericarditis, basal-cell carcinoma, and Grover's disease, presented with rapid sequential painless vision loss in both eyes (right>left) and no other neurological symptoms. At nadir, the visual acuity was no light perception (right) and 20/25 (left). Fundoscopic examination revealed bilateral optic disc

* Corresponding author at: Department of Neurology, Department of Laboratory Medicine and Pathology, Mayo Clinic, 200 First Street SW, Rochester, MN 55905, USA.

E-mail address: zekeridou.anastasia@mayo.edu (A. Zekeridou).

<https://doi.org/10.1016/j.jneuroim.2024.578312>

Received 27 October 2023; Received in revised form 22 January 2024; Accepted 4 February 2024

Available online 5 February 2024

0165-5728/© 2024 Elsevier B.V. All rights reserved.

edema (right > left). On optical coherence tomography (OCT), the peripapillary retinal nerve fiber layer (RNFL) thickness was elevated at 217 μm (right) and 195 μm (left). Brain and orbit MRI were normal. CSF analysis showed normal opening pressure (OP, 15 cm H₂O), 8 nucleated cells/ μL (91% lymphocytes), normal protein, glucose, and no oligoclonal bands. Cytology, meningitis/encephalitis panel, bacterial and fungal cultures were negative. Serum aquaporin-4 (AQP-4) and myelin oligodendrocyte glycoprotein (MOG) antibodies were negative. IgLON5-IgG was positive in both serum (1:15360) and CSF (1:16). HLA genotyping was not performed. He was treated with intravenous (IV) methylprednisolone 1000 mg daily for 5 days followed by 7 sessions of therapeutic plasma exchange and intravenous immunoglobulin (IVIG) 1 g/kg divided over 3 days, continued at 0.4 g/kg every 3 weeks as maintenance. Upon follow-up 2 months later, his vision improved from no light perception to counting fingers (right) and 20/20 (left) with an improved color vision on the left (Ishihara plate from 2/14 to 13/14). On fundoscopic exam, there was disc pallor in the right eye and chronic grade 3 disc edema on the left. (Fig. 1A). OCT showed RNFL layer thickness of 85 μm (right) and 184 μm (left). There were no abnormalities of the macula to suggest neuroretinitis. (Fig. 1D) The fluorescein angiogram was normal other than leakage from the left optic disc. A thyroid nodule was incidentally noted and subsequently proven to be metastatic medullary thyroid carcinoma.

3.2. Patient 2

A man in his late 50s developed episodes of transient binocular diplopia and pressure-like sensation in his head. He was evaluated by an ophthalmologist and had grade 2 disc edema in both eyes, bilateral superior arcuate visual field loss (Fig. 1F), and right abduction deficit. Initial CSF analysis had normal OP (20 cm H₂O), 17 nucleated cells (95% lymphocytes), elevated protein at 86 mg/dL, and normal glucose. MRI/MR venogram was unremarkable but he was treated with acetazolamide for presumed papilledema despite the normal OP. He was evaluated in our institution 6 months later: optic disc edema had resolved, and superior arcuate defects were stable in both eyes. On OCT, average RNFL thickness decreased from 132 μm to 76 μm (right) and from 135 μm to 70 μm (left). (Fig. 1E) There were no abnormalities of the macula on OCT. In the context of normal OP, the presence of pleocytosis, and significant superior arcuate defects in the setting of only grade 2 (mild) optic disc edema that would not normally cause vision loss, papillitis was favored over papilledema from intracranial hypertension; acetazolamide was discontinued. The patient also complained of gait imbalance with a reassuring exam and insomnia. A repeat lumbar puncture 1 year after symptom onset revealed a normal OP of 24 cm H₂O, persistent pleocytosis (7 nucleated cells, 88% lymphocytes), and elevated protein of 93 mg/dL. IgLON5-IgG was positive (1:32). HLA genotyping was not performed. There was no recurrence of the papilledema off acetazolamide. A sleep study was notable for rapid eye movement (REM) sleep without atonia. Neuropsychometric test showed mild cognitive deficits in executive functions, visual memory, and learning. He was treated with IV methylprednisolone 1 g for 3 days, followed by 1 g weekly for 11 weeks with a repeat CSF analysis after showing resolved pleocytosis but elevated OP at 39 cm H₂O, with no recurrence of optic disc edema and stable visual field defects. Due to symptom resolution, no further immunotherapy was pursued and the patient was stable one year later.

4. Discussion

We describe two cases of papillitis associated with IgLON5 autoimmunity, both presenting subacutely and with lymphocytic pleocytosis in the CSF. In the first case, papillitis was the sole clinical manifestation, while the second patient had some minor cognitive and gait symptoms with a reassuring exam. While non-arteritic anterior ischemic optic neuropathy (NAION) is the most common cause of optic neuropathy in

elderly patients, the spinal fluid evaluation showed pleocytosis in both patients that would not be compatible with NAION; the bilateral simultaneous involvement would also argue against this diagnosis. In addition, the first patient presented with no light perception vision in one eye that improved to counting fingers with immunotherapy, which would be atypical for NAION. The second patient presented with bilateral superior arcuate visual field deficits which would be out of proportion to the mild optic disc edema if this were papilledema from raised intracranial pressure. Further complicating the course was the elevated OP detected on the subsequent lumbar puncture (normal when the patient was initially symptomatic), however, the optic disc edema was no longer present arguing against the correlation between the two.

IgLON5-IgG detection is based on the characteristic staining on murine tissue indirect immunofluorescence and the positivity on a cell-based assay making this antibody highly specific when properly tested, as in our patients; multiple control patients with recurrent papillitis phenotype (in this study), as well as IgLON5 autoimmunity mimics (evaluated previously), were all negative (Honorat et al., 2017). Although alternative causes for papillitis such as unidentified antibodies and non-humoral inflammation are possible, IgLON5 autoimmunity remains most likely given the presence of some mild symptoms (gait, sleep disturbances) consistent with IgLON5 autoimmunity and the similarities between the two cases. Interestingly, the first patient was found to have medullary thyroid carcinoma as a part of the systemic evaluation. Paraneoplastic optic neuropathies have been well-reported, particularly in neuroendocrine tumor patients (Thirkill et al., 1989; Cross et al., 2003; Slamovits et al., 2013). However, paraneoplastic IgLON5 autoimmunity is uncommon and IgLON5 expression was not investigated in the tumor to prove a potential paraneoplastic phenomenon (Gaig et al., 2017; Graus et al., 2021; Grüter et al., 2023; Honorat et al., 2017).

The optic nerve involvement seen in these two cases was papillitis, as the optic nerve head seemed to be preferentially affected without signal abnormalities or enhancement in other segments of the optic nerve on the orbit MRI. IgLON5 mRNA transcripts were found in the ganglion cell layer of the retina in a developing mouse, which may explain the involvement of the optic nerve head (Fearnley et al., 2021). Therefore, we suspect that papillitis could be a unique presentation of IgLON5 autoimmunity, as seen for example in patients with glial fibrillary acidic protein or collapsin response mediator protein-5 autoimmunity (Cohen et al., 2020; Chen et al., 2018). Recently another patient was briefly described with IgLON5 autoimmunity and bilateral optic disc edema followed by sleep disorders and orthostatic intolerance, which further supports that papillitis could potentially be a unique presentation in this disease (Varma-Doyle et al., 2023). Future studies are needed to characterize IgLON5 expression and mechanisms of autoimmunity in patients with this rare ocular presentation. In addition, another case of cranial nerve (trigeminal) involvement was described recently in IgLON5 autoimmunity (Cluse et al., 2023) raising the question of the spectrum of IgLON5 autoimmunity.

5. Conclusions

Our study suggests that the optic nerve can rarely be involved in IgLON5 autoimmunity. Checking for IgLON5 antibodies could be considered in patients with papillitis of unclear etiology, especially in elderly patients with or without accompanying cognitive, sleep, and movement disorders. Early immunotherapy in patients with IgLON5 autoimmunity is associated with better outcomes hence early recognition of this disease is critical (Grüter et al., 2023). Larger scale studies are needed to further confirm and characterize this rare phenotype.

Funding

None.

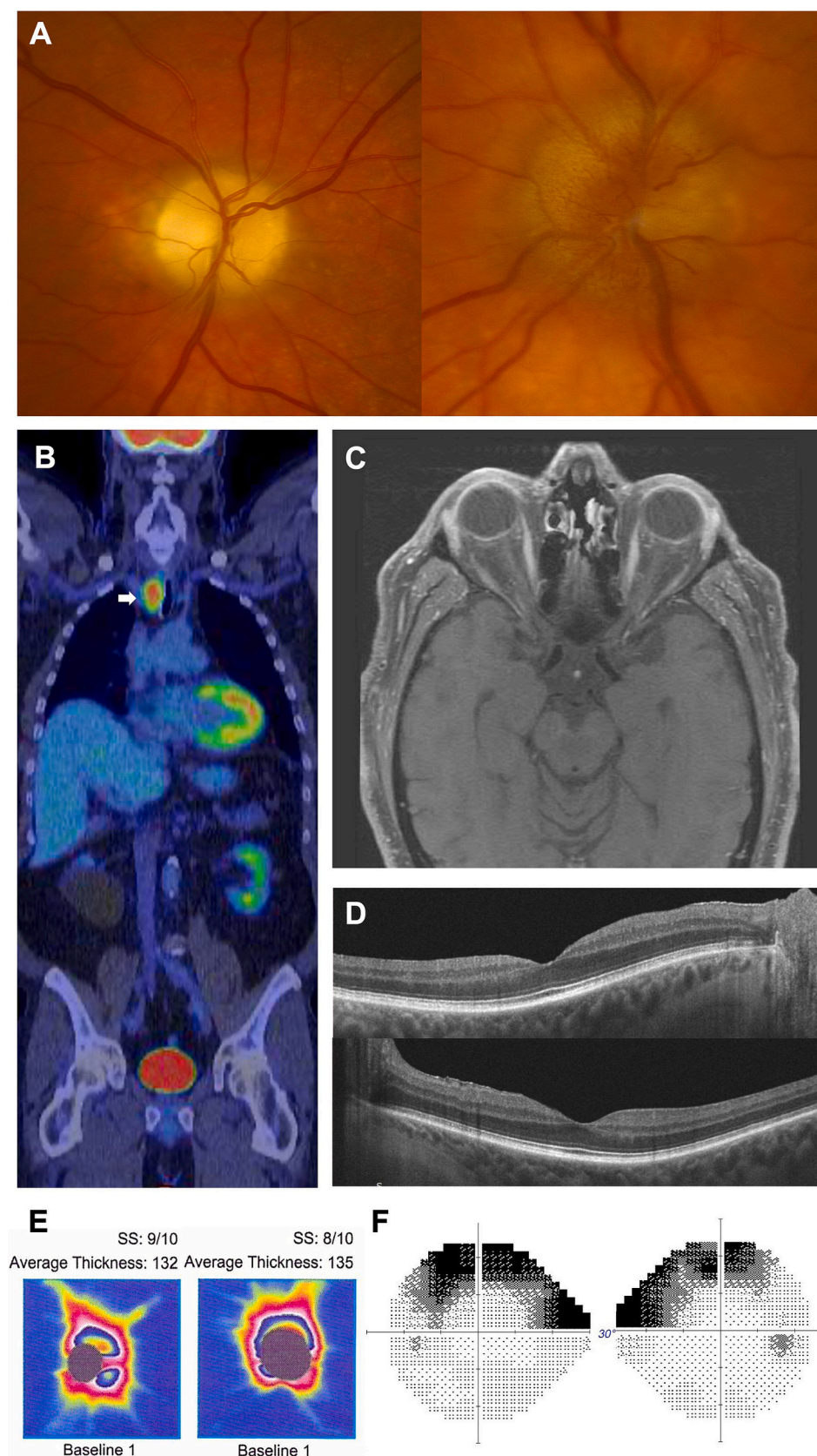


Fig. 1. (A-D, Patient 1): A. Funduscopy examination showed optic disc pallor in the right eye and grade 3 optic disc edema in the left eye 2 months after the onset of vision loss; B. PET-CT showed intensely FDG avid superior mediastinal/right paratracheal mass (*arrow*); C. MRI orbits T1 post-contrast axial image showed no optic nerve enhancement. D. OCT macular cuts demonstrated intact retinal structures in both eyes with no signs of neuroretinitis. (E-F, Patient 2): E. OCT demonstrated bilateral optic disc edema with average RNFL of 132 μ m in the right eye and 135 μ m in the left eye. F. Visual field testing demonstrated bilateral superior arcuate visual field defects in both eyes stable at the last follow-up (>2 years).

CRediT authorship contribution statement

Xiaoyang Li: Methodology, Writing – original draft, Writing – review & editing, Data curation. **John J. Chen:** Formal analysis, Writing – review & editing. **Minjun Hur:** Formal analysis, Writing – review & editing. **Gillian R. Paton:** Formal analysis, Writing – review & editing. **Andrew McKeon:** Conceptualization, Formal analysis, Writing – review & editing. **Anastasia Zekeridou:** Conceptualization, Formal analysis, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

John J. Chen serves as a consultant for UCB and Horizon; Andrew McKeon reports research funding from the National Institutes of Health (NIH: RO1NS126227, U01NS120901), patents issued for GFAP and MAP1B-IgGs and patents pending for PDE10A, Septins-5 and -7, and KLCHL11-IgGs, and has consulted for Janssen and Roche pharmaceuticals, without personal compensation; Anastasia Zekeridou reports patents pending for PDE10A-IgG, Tenascin-R-IgG and DACH1-IgG as biomarkers of neurological autoimmunity; research funding from Roche not relevant to the current project; consulting for Alexion Pharmaceuticals with no personal compensation.

Data availability

Data will be made available on request.

Acknowledgments

We thank the patients and their families for their participation in this study. We would also like to thank Lilian Mills for performing the antibody testing in the control patients.

References

- Chen, J.J., Aksamit, A.J., McKeon, A., Pittock, S.J., Weinshenker, B.G., Leavitt, J.A., et al., 2018. Optic disc edema in glial fibrillary acidic protein autoantibody-positive meningoencephalitis. *J. Neuroophthalmol.* 38 (3), 276–281.

- Cluse, F., Hermier, M., Demarquay, G., Rogemond, V., Mallaret, M., Svahn, J., et al., 2023. Trigeminal nerve involvement in bulbar-onset anti-IgLON5 disease. *Neurol. Neuroimmunol. Neuroinflamm.* 10 (6).
- Cohen, D.A., Bhatti, M.T., Pulido, J.S., Lennon, V.A., Dubey, D., Flanagan, E.P., et al., 2020. Collapsin response-mediator protein 5-associated retinitis, vitritis, and optic disc edema. *Ophthalmology* 127 (2), 221–229.
- Cross, S.A., Salomao, D.R., Parisi, J.E., Kryzer, T.J., Bradley, E.A., Mines, J.A., et al., 2003. Paraneoplastic autoimmune optic neuritis with retinitis defined by CRMP-5-IgG. *Ann. Neurol.* 54 (1), 38–50.
- Fearnley, S., Raja, R., Cloutier, J.F., 2021. Spatiotemporal expression of IgLON family members in the developing mouse nervous system. *Sci. Rep.* 11 (1), 19536.
- Gaig, C., Graus, F., Compta, Y., Högl, B., Bataller, L., Brüggemann, N., et al., 2017. Clinical manifestations of the anti-IgLON5 disease. *Neurology* 88 (18), 1736–1743.
- Gelpi, E., Höftberger, R., Graus, F., Ling, H., Holton, J.L., Dawson, T., et al., 2016. Neuropathological criteria of anti-IgLON5-related tauopathy. *Acta Neuropathol.* 132, 531–543.
- Graus, F., Vogrig, A., Muñoz-Castrillo, S., Antoine, J.C.G., Desestret, V., Dubey, D., et al., 2021. Updated diagnostic criteria for paraneoplastic neurologic syndromes. *Neurol. Neuroimmunol. Neuroinflamm.* 8 (4).
- Grüter, T., Möllers, F.E., Tietz, A., Dargviniene, J., Melzer, N., Heidbreder, A., et al., 2023. Clinical, serological and genetic predictors of response to immunotherapy in anti-IgLON5 disease. *Brain* 146 (2), 600–611.
- Honorat, J.A., Komorowski, L., Josephs, K.A., Fechner, K., St Louis, E.K., Hinson, S.R., et al., 2017. IgLON5 antibody: neurological accompaniments and outcomes in 20 patients. *Neurol. Neuroimmunol. Neuroinflamm.* 4 (5).
- Sabater, L., Gaig, C., Gelpi, E., Bataller, L., Lewerenz, J., Torres-Vega, E., et al., 2014. A novel non-rapid-eye movement and rapid-eye-movement parasomnia with sleep breathing disorder associated with antibodies to IgLON5: a case series, characterisation of the antigen, and post-mortem study. *Lancet Neurol.* 13 (6), 575–586.
- Sista, S.R., Crum, B., Aboseif, A., Devine, M.F., Zekeridou, A., Hammami, M.B., et al., 2022. Motor-neuron-disease-like phenotype associated with IgLON5 disease. *J. Neurol.* 269 (11), 6139–6144.
- Slamovits, T.L., Posner, J.B., Reidy, D.L., Thirkill, C.E., Keltner, J.L., 2013. Pancreatic neuroendocrine paraneoplastic optic neuropathy: confirmation with antibody to optic nerve and hepatic metastasis. *J. Neuroophthalmol.* 33 (1), 21–25.
- Thirkill, C.E., FitzGerald, P., Sergott, R.C., Roth, A.M., Tyler, N.K., Keltner, J.L., 1989. Cancer-associated retinopathy (CAR syndrome) with antibodies reacting with retinal, optic-nerve, and cancer cells. *N. Engl. J. Med.* 321 (23), 1589–1594.
- Varma-Doyle, A., Chwalisz, B., Linnoila, J., 2023. Anti-IgLON-5 associated neurological disease presenting with bilateral optic neuritis, orthostatic intolerance, sleep-related movements in setting of testicular mass: expanding clinical phenotypes in the disease. In: American Academy of Neurology Annual Meeting(P5-5.028).